Kinetic energy and net work

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect net work to a change in kinetic energy.

SUCCESS CRITERIA - I can...

1. Calculate kinetic energy with speed squared.
2. Use net work rather than one force's work.
3. Connect the sign of work to a speed change.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

At fixed momentum magnitude, mass becomes 3 times as large. Derive the kinetic-energy multiplier using $K=p^2/(2m)$.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $K = \frac{1}{2}mv^2$; $W_{net} = K_f - K_i$.
i = initial; f = final. Positive net work increases kinetic energy.

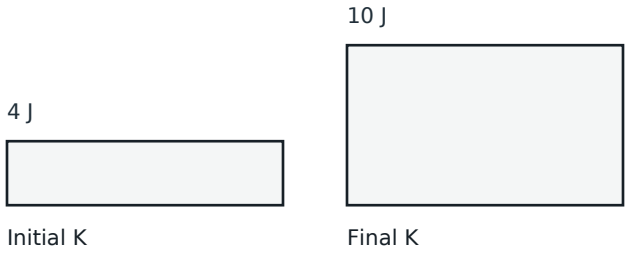
Worked reasoning: A 2 kg cart at 2 m/s has K=4 J. Adding 5 J of net work gives final K=9 J, so final speed is 3 m/s.

Your turn: Draw before-and-after energy bars and label the transfer between them.

Marzano check - circle: ① ② ③

03 Represent and interpret

Energy bars show kinetic energy changing from 7 J to 10 J. Find net work and describe the speed change for constant mass.



Bar height represents energy. Read the value and unit above each bar.

Marzano check - circle: ① ② ③

04 Solve and justify

A 4 kg cart moves at 2 m/s. Net work is 24 J. Find final kinetic energy and speed.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Analyze the reasoning

A student adds the work of a pull but ignores friction. Why might their predicted speed be too high?

Marzano check - circle: 1 2 3

06 Choose a method

Two equal 4 kg carts collide elastically along one line. A starts at +6 m/s; B starts at rest. Find both final velocities under the ideal model.

Marzano check - circle: 1 2 3

07 Exit check

A 2 kg object slows from 6 to 2 m/s. Find net work.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A 3 kg object starts at 4 m/s and receives -18 J of net work. Find final speed. What is the most negative net work possible before it first stops?

Marzano check - circle: 1 2 3